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Third-Year Undergraduate Programme (LMD)
Radiation Physics
Course: Nuclear Physics

Practical Work N° 1
Simulation and Analysis of Rutherford Scattering of Alpha Particles

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1 Introduction

In 1909, Geiger and Marsden, under Rutherford's direction, directed α particles at a thin gold foil. Thomson's model predicted only small deflections, yet most particles passed straight through, while a few were scattered at large angles and some were even backscattered. Rutherford concluded in 1911 that the positive charge and almost all the mass of the atom are concentrated in a tiny, dense nucleus. The scattering is governed by the Coulomb repulsion between the α particle (charge $2e$) and the nucleus (charge Ze). In this practical work, we use a simulation to determine the minimum distance of approach $d_{\min}$, show that it varies linearly with Z, recover the kinetic energy E_c of the α particles, and compare the simulated $\theta(b)$ with Rutherford's formula.

2 Simulation of the Experiment

The simulation was performed for alpha particles interacting with a gold nucleus ($Z = 79$), with a kinetic energy

$$E = 6\,\text{MeV}.$$

Figure 1 shows the simulation interface, displaying the scattering trajectories of alpha particles and the curve θ as a function of the impact parameter b. The minimum distance of approach obtained from the simulation is $d_{\min} = 37.92\,\text{fm}$.

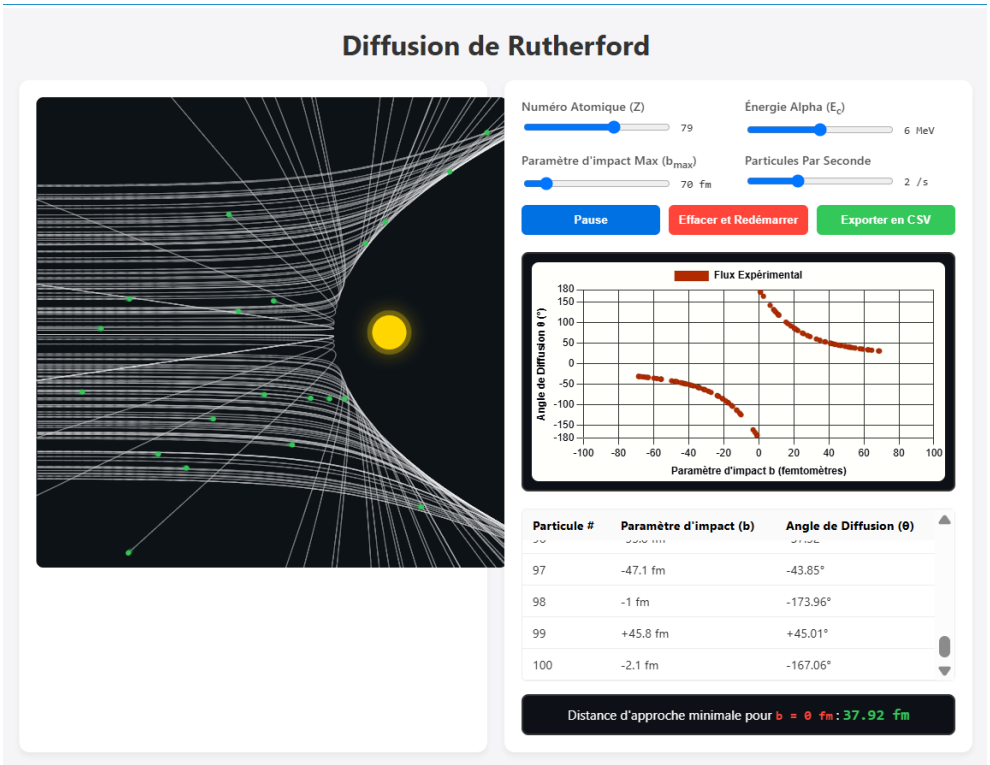


Figure 1: Simulation interface for gold ($Z = 79$) with $E = 6\,\text{MeV}$, showing scattering trajectories and the minimum distance $d_{\min}$.

The simulation was performed using the online Rutherford scattering tool, which is available here.

3 Minimum Distance of Approach

For a head-on collision, the alpha particle approaches the target nucleus until its initial kinetic energy is entirely converted into electrostatic potential energy. At the minimum distance of approach $d_{\min}$, the kinetic energy is momentarily zero.

By conservation of energy:

$$E_c = \frac{k(2e)(Ze)}{d_{\min}}. \tag{1}$$

Therefore,

$$E_c = \frac{2kZe^2}{d_{\min}}. \tag{2}$$

Rearranging this expression gives:

$$d_{\min} = \frac{2ke^2}{E_c}\, Z. \tag{3}$$

Defining the constant

$$C_\alpha = \frac{2ke^2}{E_c}, \tag{4}$$

we obtain the required linear relationship:

$$\boxed{d_{\min} = C_\alpha Z}. \tag{5}$$

Thus, for a fixed kinetic energy, the minimum distance of approach is directly proportional to the atomic number Z of the target nucleus.

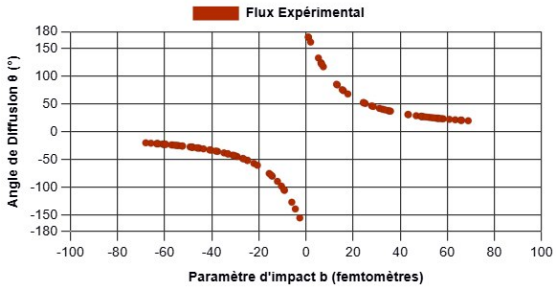
4 Calculation of the Kinetic Energy E_c of the Alpha Particles

The alpha-particle kinetic energy was kept constant at $E = 6\,\text{MeV}$. Simulations were performed for four atomic numbers, $Z = 50, 60, 70$, and 80. For each value of Z, the minimum distance of approach, $d_{\min}$, was recorded from the simulator's output field (the simulator does not display it on the plotted curve) and is reported in Table 1. The corresponding simulation results are presented in Figure 2.

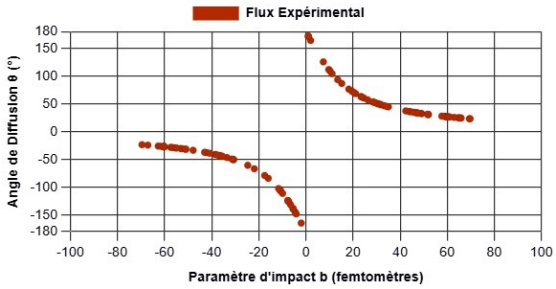
Z	d_m simulated (fm)	d_m theoretical (fm)	Relative error (%)
50	24.00	24.00	0
60	28.80	28.80	0
70	33.60	33.60	0
80	38.40	38.40	0

Table 1: Minimum distance of approach d_m for different values of Z ($E = 6\,\text{MeV}$). The theoretical values are computed from $d_m = \dfrac{2ke^2}{E_c}\, Z = \dfrac{2 \times 1.44}{6}\, Z$ with $ke^2 = 1.44\,\text{MeV fm}$.

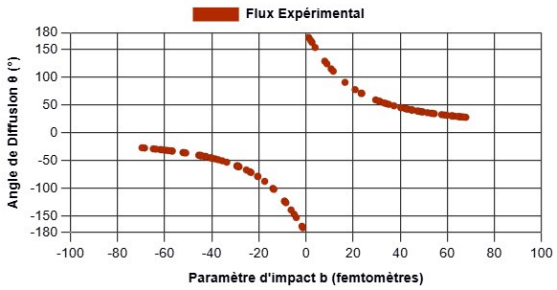
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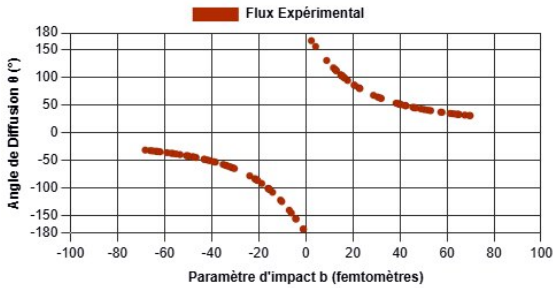
(a) $Z = 50$ ($d_{\min} = 24.00$ fm)



(b) $Z = 60$ ($d_{\min} = 28.80$ fm)



(c) $Z = 70$ ($d_{\min} = 33.60$ fm)



(d) $Z = 80$ ($d_{\min} = 38.40$ fm)

Figure 2: Simulation curves for $Z = 50, 60, 70, 80$ at $E = 6\,\text{MeV}$.

The experimental data are plotted to verify the linear relationship between $d_{\min}$ and Z.

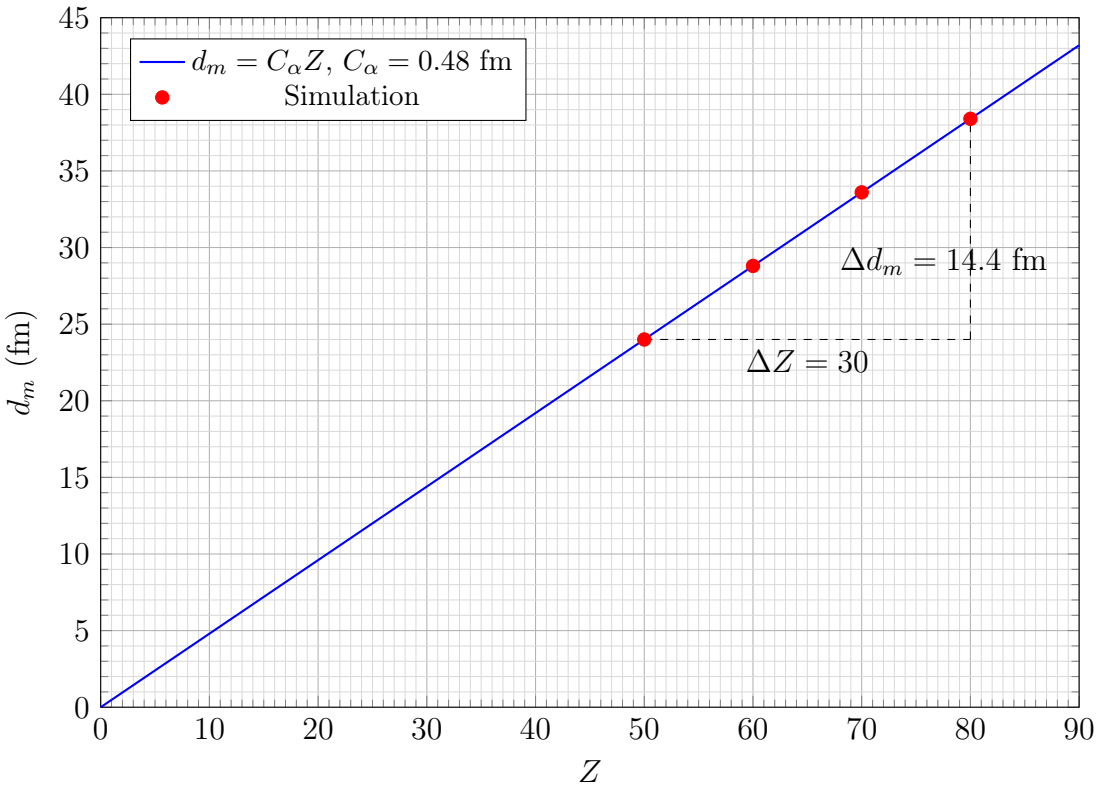


Figure 3: Minimum distance of approach d_m versus atomic number Z ($E = 6\,\text{MeV}$). The slope of the line gives C_α.

The slope of the line in Figure 3 is

$$C_\alpha = \frac{\Delta d_m}{\Delta Z} = \frac{38.40 - 24.00}{80 - 50} = \frac{14.40}{30} = 0.48 \text{ fm}. \tag{6}$$

$$E_c = \frac{2ke^2}{C_\alpha} = \frac{2 \times 1.44}{0.48} = 6.00 \text{ MeV}. \tag{7}$$

Note: Using $k = 8.99 \times 10^9$ N $\cdot$ m^2/C^2 and $e = 1.60 \times 10^{-19}$ C:

$$ke^2 = \frac{(8.99 \times 10^9) \times (1.60 \times 10^{-19})^2}{(1.602 \times 10^{-13} \text{ J/MeV}) \times (10^{-15} \text{ m/fm})} \approx 1.44 \text{ MeV} \cdot \text{fm}$$

The relative error with respect to the imposed energy (6 MeV) is 0 %, which confirms the linear law $d_m = C_\alpha Z$.

5 Scattering Angle

Method 1: Plot with Python (Matplotlib)

The scattering angle θ depends on the impact parameter b, the atomic number Z of the target nucleus, and the initial kinetic energy E_c of the alpha particles. According to Rutherford's scattering theory, the theoretical scattering angle is given by

$$\theta(b) = \frac{180}{\pi} 2 \arctan\left(\frac{ke^2 Z}{E_c b}\right), \tag{8}$$

where $ke^2 = 1.44$ MeV fm, E_c is the kinetic energy of the alpha particles, and b is the impact parameter expressed in femtometers (fm). The angle θ is expressed in degrees.

For a gold nucleus, the parameters are

$$Z = 79, \qquad E_c = 6 \text{ MeV}.$$

The simulation data are exported as a CSV file containing the 100 pairs (b, θ). These data are plotted together with the theoretical curve given by Equation (8) in Figure 4.

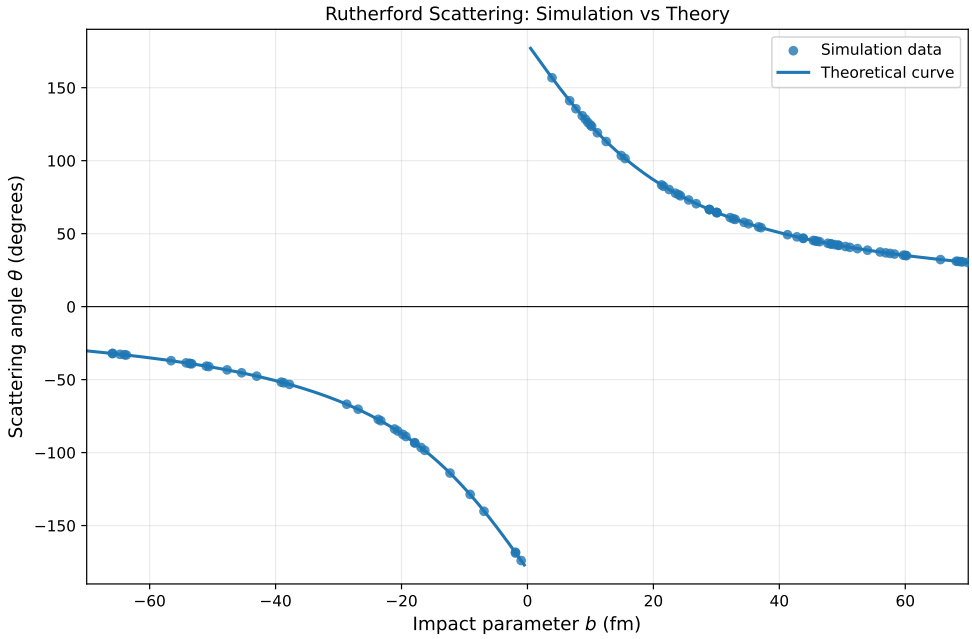


Figure 4: Comparison between the simulated and theoretical Rutherford scattering angles for gold ($Z = 79$) at $E_c = 6\,\text{MeV}$.

The comparison allows us to assess the agreement between the simulation results and Rutherford's theoretical prediction. The scattering angle decreases as the impact parameter increases: small values of b (near head-on collisions) give large deflection angles, while large values of b give almost undeviated trajectories.

Python Code for the Scattering Angle

The following script processes the exported simulation data and plots it against Rutherford's theoretical formula:

```
1
2  import numpy as np
3  import pandas as pd
4  import matplotlib.pyplot as plt
5  from pathlib import Path
6
7  # ------------------------------------------------------
8  # 1. Read the simulation data
9  # ------------------------------------------------------
10 data = pd.read_csv("data.csv")
11
12 b = data["Impact_Parameter_b_fm"].to_numpy()
13 theta_sim = data["Diffusion_Theta_Degrees"].to_numpy()
14
15 # ------------------------------------------------------
16 # 2. Define the theoretical Rutherford scattering law
17 # Z = 79, Ec = 6 MeV, k*e^2 = 1.44 MeV.fm
18 # ------------------------------------------------------
19 Z = 79
20 Ec = 6.0
```

```
ke2 = 1.44

def theta_theory(b):
    return np.degrees(
        2 * np.arctan((ke2 * Z) / (Ec * b))
    )

# Split the curve at b = 0 to avoid division by zero
b_negative = np.linspace(-70, -0.5, 1000)
b_positive = np.linspace(0.5, 70, 1000)

# ----------------------------------------------------
# 3. Plot simulation points and theoretical curves
# ----------------------------------------------------
fig, ax = plt.subplots(figsize=(9, 6))

ax.scatter(
    b, theta_sim,
    s=28,
    alpha=0.8,
    label="Simulation data",
    zorder=3
)

line, = ax.plot(
    b_negative,
    theta_theory(b_negative),
    linewidth=2,
    label="Theoretical curve"
)

ax.plot(
    b_positive,
    theta_theory(b_positive),
    linewidth=2,
    color=line.get_color()
)

ax.set_xlabel(r"Impact parameter $b$ (fm)", fontsize=12)
ax.set_ylabel(r"Scattering angle $\theta$ (degrees)",
   fontsize=12)

ax.set_title("Rutherford Scattering: Simulation vs Theory")
ax.set_xlim(-70, 70)
ax.set_ylim(-190, 190)

ax.axhline(0, color="black", linewidth=0.7)
ax.grid(True, alpha=0.25)
ax.legend()
fig.tight_layout()

```

```python
# ----------------------------------------------------
# 4. Save the figures
# ----------------------------------------------------
output_dir = Path("images/graphes")
output_dir.mkdir(parents=True, exist_ok=True)

fig.savefig(
    output_dir / "rutherford_scattering.png",
    dpi=300,
    bbox_inches="tight"
)

fig.savefig(
    output_dir / "rutherford_scattering.pdf",
    bbox_inches="tight"
)

print(f"Number of simulation points: {len(data)}")
print("Figures saved successfully in the images/graphes
   folder.")

plt.show()
```

Listing 1: Python script for processing and plotting Rutherford scattering data.

Method 2: Plot with TikZ (PGFPlots)

The same data are plotted again using TikZ for a publication-quality vector figure.

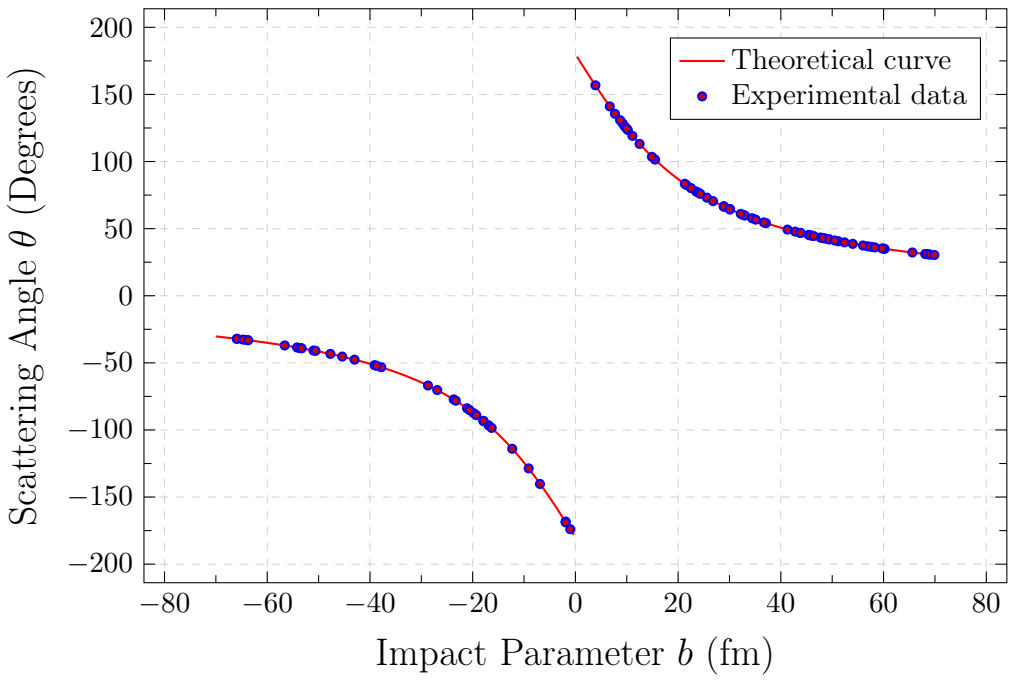


Figure 5: Relationship between impact parameter and scattering angle.

6 Conclusion

This practical work demonstrated the linear dependence of the minimum distance of approach ($d_{\min}$) on the atomic number (Z), allowing the initial kinetic energy of the alpha particles to be recovered as ($E_c = 6.00$ MeV). The comparison between the simulated and theoretical scattering angles also showed excellent agreement with Rutherford's scattering law for gold. These results illustrate how numerical simulation can be used to test theoretical predictions in nuclear physics.

7 Source Code and Reproducibility

To ensure transparency and reproducibility, all Python source code, CSV data files, TikZ figures, graphics, and the full LaTeX source code for this document are publicly available on my GitHub repository. `GitHub Repository`

Scan or click to access the repository

8 Interactive Web Simulation

As an addition, I recreated the Rutherford scattering simulation using **JavaScript** with AI assistance, and published it on our **Physics Club** website.

Try the Web Simulation →

Scan or click to open the simulation

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